

Homoscedasticity and Its Testing: From Conditional Expectation to the Breusch–Pagan Test

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Part 2 / 12
Likelihood, Score, and Fisher Information

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1 Density and Likelihood

This chapter develops two closely linked objects. The first half treats the parametric family of densities $\{f_\theta\}_{\theta \in \Theta}$ as a single bivariate function $h(x, \theta) = f_\theta(x)$: a deterministic surface, with no data attached. We collect the properties of h — normalisation in x , and the vanishing θ -integral of $\partial_\theta h$ — and note how the latter follows from the former by three successive rephrasings of one trivial fact. The second half acknowledges that observations are random and slices h in the θ direction. Depending on whether the slice index is a number x or the random variable $X(\omega)$, the object is called the **pre-likelihood** or the **likelihood** proper. Both readings live on the same surface; which one applies depends on what is plugged into the first slot.

We follow the Probabilist's notes of May 10, 2026 closely throughout.

1.1 The joint function $h(x, \theta)$ and its two readings

Setup. Fix a measurable space $(\Omega, \mathcal{F})$ and a parametric family of probability measures $\{P_\theta\}_{\theta \in \Theta}$ on it. Assume every P_θ has a density $f_\theta(x)$ with respect to some common dominating measure (e.g. Lebesgue or counting measure on $\mathbb{R}$).

There is a single joint object here:

$$h(x, \theta) = f_\theta(x) = L_x(\theta).$$

The same real number $h(x, \theta)$ is a density *in x* (when θ is held fixed) and a pre-likelihood *in θ* (when x is held fixed).

The surface and its slices. The Probabilist's diagram (May 10, 2026, page 1) draws h as a surface over the (x, θ) plane, with x pointing right, θ pointing left/forward, and h rising vertically. Two natural families of slices live on it:

- **Density slices** (fixed θ , varying x): $h(\cdot, \theta_0) = f_{\theta_0}$ is a probability density in x .
- **Pre-likelihood slices** (fixed x , varying θ): $h(x_1, \cdot) = L_{x_1}$ is the pre-likelihood as a function of θ (indexed by the *number* x_1). When x_1 is later replaced by the random $X(\omega)$, this slice becomes the **likelihood** $L(\omega, \theta)$; see Section 1.4.

Figure 1 shows both kinds of slices on the same surface for the location-normal model $\mathcal{N}(\theta, 1)$.

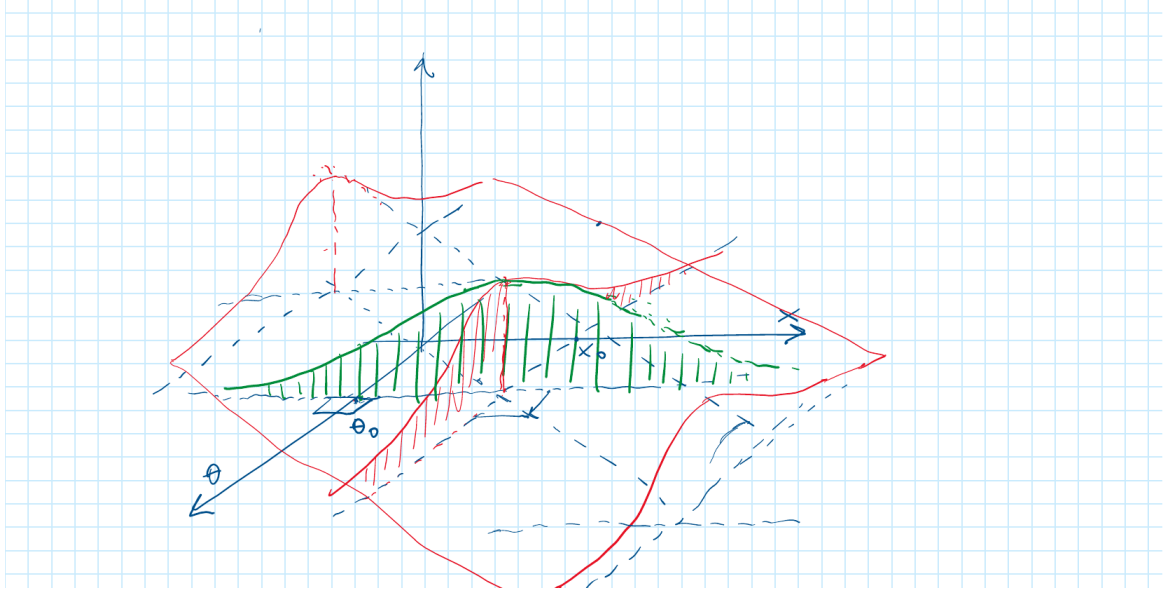


Figure 1: Bivariate surface $h(x, \theta) = f_\theta(x) = L_{x_1}(\theta)$. Vertical axis: h . Horizontal axes: x (right) and θ (forward-left). The **green** curve runs along the x -axis at a fixed value θ_0 : this is the density slice $f_{\theta_0}(x)$, and the green hatching shows its vertical curtain dropping to the floor. The **red** hatching near x_1 runs along the θ -axis at a fixed *number* x_1 : this is the pre-likelihood slice $L_{x_1}(\theta)$ (when x_1 is later replaced by $X(\omega)$, this slice becomes a random function of θ — the likelihood proper; see Section 1.4). Both slices live on the same surface, and for the location-normal family both happen to look bell-shaped (see Remark 1).

1.2 Three rephrasings: from $\int f_\theta dx = 1$ to $\int \partial_\theta h dx = 0$

The fact that every density integrates to one — a trivial axiom — turns out to have a non-trivial consequence: the θ -derivative of h integrates to zero in the data variable. We obtain this by three successive rephrasings, each a small linguistic shift. Nothing about scores or likelihoods enters yet; this is a fact about the surface h .

Step 0: the starting fact. Every f_θ is a density, so

$$\int_{-\infty}^{\infty} f_\theta(x) dx = 1 \quad \text{for every } \theta.$$

This is a *family* of univariate functions $\{f_\theta(\cdot)\}$, each integrating to the same value (namely 1):

$$\forall \theta_1, \theta_2 : \int_{-\infty}^{\infty} f_{\theta_1}(x) dx = \int_{-\infty}^{\infty} f_{\theta_2}(x) dx.$$

Rephrasing 1: family of functions $\rightarrow$ bivariate function. If a family of univariate functions all integrate to the same value, we can rephrase this as: *a bivariate function integrates over one argument to a constant*. Naming the bivariate object h :

$$\int_{-\infty}^{\infty} h(x, \theta) dx \equiv c \quad (\text{here } c = 1).$$

We have promoted the parameter θ from a family index to a second argument.

Rephrasing 2: constant $\rightarrow$ function not depending on θ . A constant in θ is, trivially, a function of θ that *does not depend on θ* . Let $A(\theta) := \int h(x, \theta) dx$. Then $A(\theta) \equiv c$, so $A'(\theta) = 0$ for every θ .

Rephrasing 3: derivative of integral $\rightarrow$ integral of derivative. If we may differentiate under the integral sign (Leibniz / dominated convergence; holds for exponential families and most standard models), then the derivative of A equals the integral of the partial derivative of h along θ :

$$A'(\theta) = \frac{d}{d\theta} \int_{-\infty}^{\infty} h(x, \theta) dx = \int_{-\infty}^{\infty} \frac{\partial h(x, \theta)}{\partial \theta} dx = 0.$$

In words: *the derivative of the bivariate function h with respect to the remaining argument θ , integrated over the first argument x , is identically zero.*

This is the property the Probabilist draws on page 2 of the notes. The θ -derivative surface $\partial_{\theta}h$ has positive regions (where the density rises as θ increases) and negative regions (where it falls); along any fixed- θ line, these regions cancel *exactly*.

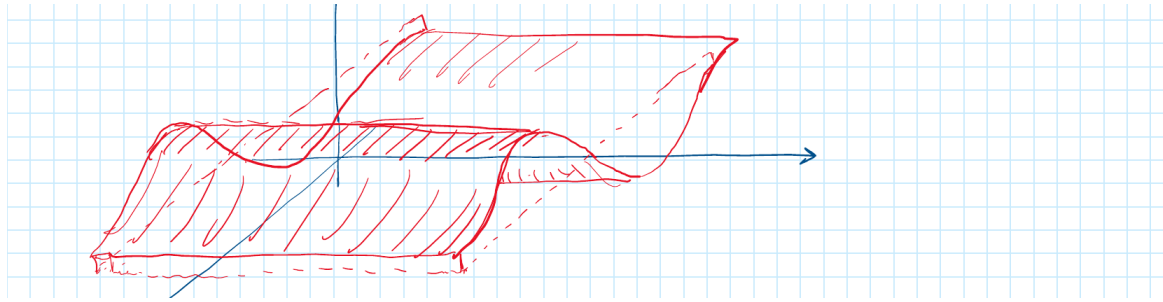


Figure 2: $\partial_{\theta}h(x, \theta) = (x - \theta)\varphi(x - \theta)$. Vertical axis: $\partial_{\theta}h$. Horizontal axes: x (right) and θ (forward-left). The surface dips below zero on one side of the $x = \theta$ ridge and rises above zero on the other. The green hatching shows the magnitude of $\partial_{\theta}h$ as a vertical curtain dropped from the surface to the θ -axis plane. At any fixed θ , the positive and negative regions have equal area: $\int \partial_{\theta}h(x, \theta) dx = 0$. The point x_1 on the x -axis is the fixed observation through which a likelihood slice would be taken.

Property summary. The three rephrasings give a chain:

$$\underbrace{\int f_{\theta}(x) dx = 1}_{\text{density axiom}} \iff \underbrace{\int h(x, \theta) dx \equiv c}_{\text{constant in } \theta} \xrightarrow{\text{Leibniz}} \underbrace{\int \partial_{\theta}h(x, \theta) dx = 0}_{\text{Property 3}}. \quad (1)$$

Remark 1 (Location-normal symmetry is a coincidence). In Figures 1 and 2 the surfaces are symmetric under swapping x and θ . This is *not* a general property of likelihoods — it is a coincidence specific to the location-normal family with $\sigma^2 = 1$, where

$$f_{\theta}(x) = \varphi_{\mu}(x) = \varphi(x - \mu) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{(x - \mu)^2}{2}\right\}.$$

Because $h(x, \theta)$ depends on x and θ only through the difference $x - \theta$, the surface h is invariant under $(x, \theta) \mapsto (\theta, x)$ followed by reflection. As a result, the likelihood slice $L_{x_1}(\theta)$ has the same shape as the density slice $f_{\theta_0}(x)$.

For a general parametric family this symmetry fails. Consider the exponential model $f_{\theta}(x) = \theta e^{-\theta x}$ on $x > 0$, $\theta > 0$. The density slice (fixed θ) is a decreasing exponential in x . The likelihood slice (fixed $x_1 > 0$), $L_{x_1}(\theta) = \theta e^{-\theta x_1}$, is *not* monotone: it rises, peaks at $\hat{\theta} = 1/x_1$, then decays. The two slices have qualitatively different shapes, and there is no reason to expect otherwise.

1.3 Cancellation in a 2D picture

The 3D surfaces are striking but hard to read precisely. Two flat panels make the same point.

Left: the family $\{f_\theta(\cdot)\}$ as overlaid bell curves. All have area 1. The means of the bells are at θ ; shifting θ slides the bell along x .

Right: the integrand $\partial_\theta h(x, \theta_0)$ at fixed $\theta_0 = 0$. For the location-normal family this equals $x \cdot \varphi(x)$ — positive for $x > 0$, negative for $x < 0$, with equal areas on the two sides.

The constant $A(\theta)$ and its zero derivative. A trivial but useful pair: $A(\theta) = \int h \, dx$ is identically 1 (left), so its derivative is identically 0 (right). These two panels are Rephrasings 1 and 3 of the chain in equation (1) drawn explicitly.

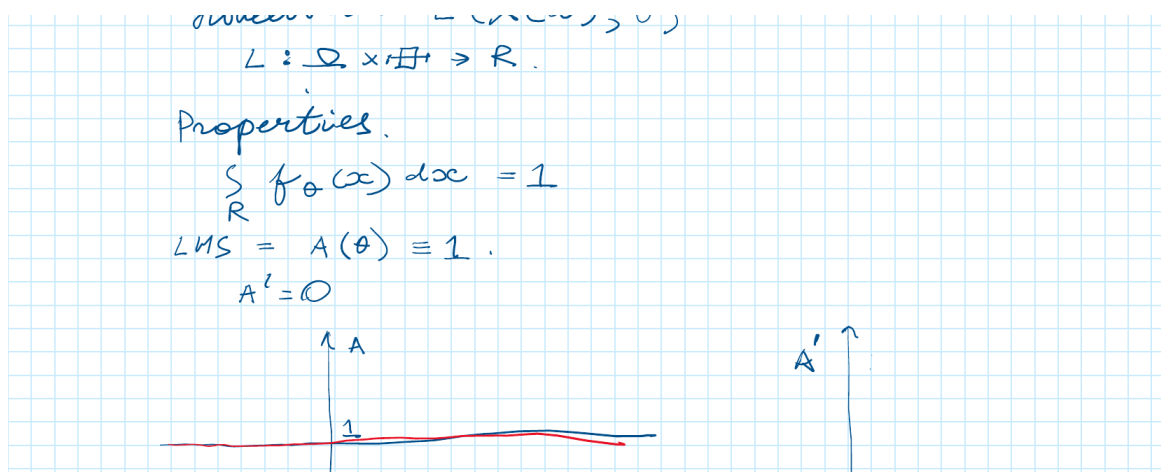


Figure 3: Left: $A(\theta) = \int h(x, \theta) \, dx \equiv 1$ — a horizontal line at height 1. Right: $A'(\theta) \equiv 0$ — a horizontal line at height 0.

The $\partial_\theta h$ surface, as a curtain. Another way to picture Property 3 is to draw $\partial_\theta h$ itself, as a curtain hanging over (and under) the data axis. At any fixed θ , the curtain is positive in some regions of x (where h is rising with θ) and negative in others (where h is falling). The shaded positive and negative regions balance: their total signed area is zero.

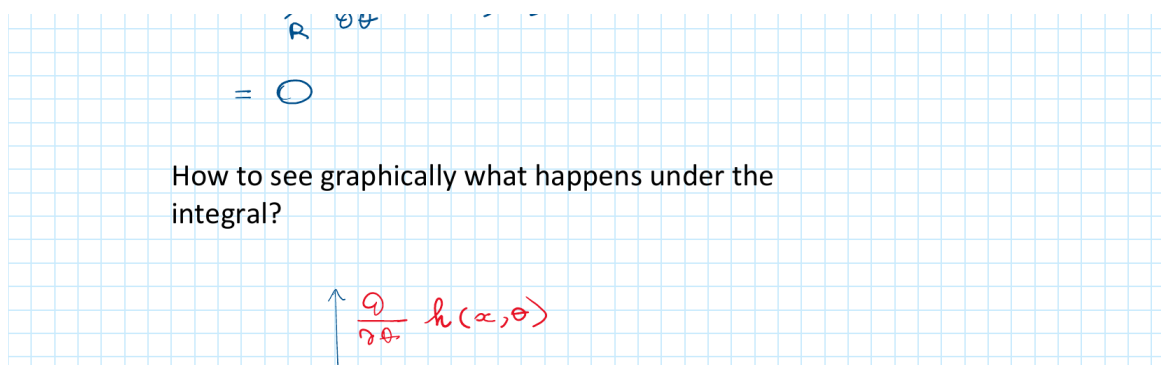


Figure 4: The curtain of $\partial_\theta h(x, \theta)$ over the x -axis. Above the zero plane: $\partial_\theta h > 0$; below: $\partial_\theta h < 0$. For any horizontal slice of the curtain (fixed θ), the positive and negative parts have equal area, so $\int \partial_\theta h \, dx = 0$.

argument, integrates to zero over the first argument.

$$\int_{-\infty}^{\infty} \frac{\partial}{\partial \theta} h(x, \theta) dx \equiv 0$$

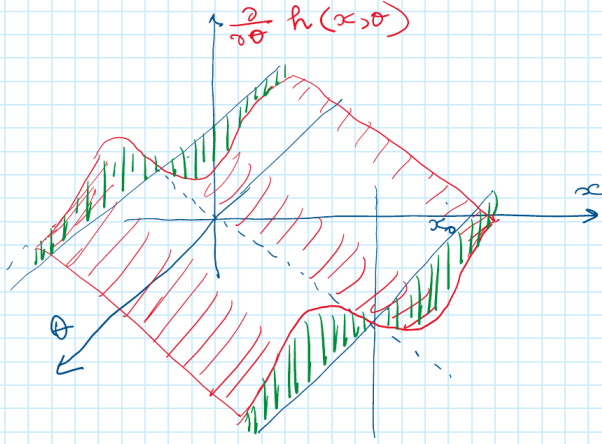
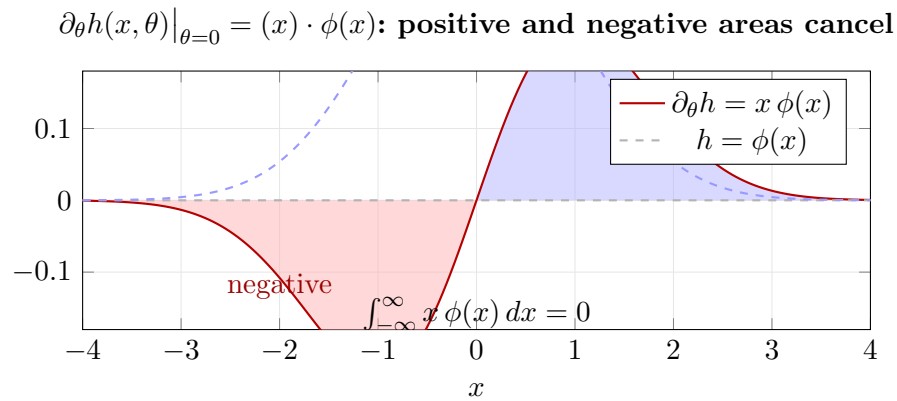


Figure 5: The Probabilist’s hand-drawn picture of $\partial_{\theta}h(x, \theta)$ as a surface over the (x, θ) plane. Vertical axis: $\partial_{\theta}h(x, \theta)$. Horizontal axes: x (right) and θ (forward-left). The red hatching shows $\partial_{\theta}h$ above and below the zero plane; the green spikes show the signed density cancellation at a fixed x_0 . At any fixed θ , the positive and negative parts integrate to zero: $\int \partial_{\theta}h(x, \theta) dx = 0$.

Why do positive and negative regions cancel? The key is that $h(x, \theta)$ is a Gaussian bell centred at θ . When θ increases, the bell shifts right. The left tail of the density falls (so $\partial_{\theta}h < 0$ there) and the right tail rises (so $\partial_{\theta}h > 0$ there). These gains and losses are symmetric and equal: the total probability is conserved.

The following inline diagram (for $\mathcal{N}(\theta, 1)$ at $\theta = 0$) makes this concrete.



The shaded areas are equal in magnitude. This is Property 3 in geometric form: the net signed area under $\partial_{\theta}h$ is zero. This fact, on its own, has consequences — but those will be drawn out in Chapter 3, once we have introduced the score.

1.4 Pre-likelihood vs. likelihood: numeric vs. random argument

So far the surface h and its properties have involved no data: h is a fixed bivariate function and all of its slices are deterministic curves. Once we acknowledge that observations are random, the same surface gives rise to two different objects depending on what we plug into its first slot — a number,

or a random variable. This is the distinction the Probabilist insists on, and it sets up everything in the next two chapters: the random nature of the log-likelihood (Chapter 2), and the zero-mean property of the score (Chapter 3).

So far we have treated x as a deterministic argument — a coordinate along an axis. Once we acknowledge that the observation is, formally, a random variable $X : \Omega \rightarrow \mathbb{R}$, two different objects split apart. The Probabilist’s terminology, which we follow here, draws the line by *what is plugged into the first slot of h* .

Pre-likelihood: first slot is a number. Fix a *number* $x \in \mathbb{R}$, i.e. a point on the axis, with no probabilistic structure attached. The slice of h at this point is

$$L_x : \Theta \rightarrow \mathbb{R}, \quad L_x(\theta) := h(x, \theta) = f_\theta(x).$$

This is a deterministic function of θ , indexed by a real number x . The Probabilist calls it the **pre-likelihood**. The graph of L_x in the (θ, h) plane is a single, non-random curve.

Likelihood: first slot is a random variable. Now replace the number x by $X(\omega)$ — the random variable itself, evaluated at a sample point $\omega \in \Omega$. The result is a function of two arguments:

$$L : \Omega \times \Theta \rightarrow \mathbb{R}, \quad L(\omega, \theta) := f_\theta(X(\omega)) = L_{X(\omega)}(\theta).$$

The Probabilist calls this the **likelihood** (proper). Indexed by ω , it is a *random* function of θ : each draw of ω gives a different curve $\theta \mapsto L(\omega, \theta)$.

What “pre” means here. The prefix *pre* refers to “before substituting the random variable”, not to “before taking the logarithm or maximising”. Once we plug in the random observation, the object jumps from deterministic to random:

$$\underbrace{L_x(\theta)}_{\text{indexed by number } x} \xrightarrow{x \leftarrow X(\omega)} \underbrace{L(\omega, \theta) = L_{X(\omega)}(\theta)}_{\text{indexed by } \omega}.$$

The map $\omega \mapsto L_{X(\omega)}$ is a random element of the function space of curves $\Theta \rightarrow \mathbb{R}$.

Remark 2 (A personal note). This precise distinction is the Probabilist’s preferred terminology and we adopt it throughout the chapter. Many textbooks call L_x itself “the likelihood” without making the ω vs. x split explicit; the present notation makes the measurability and randomness questions that follow much cleaner to state.

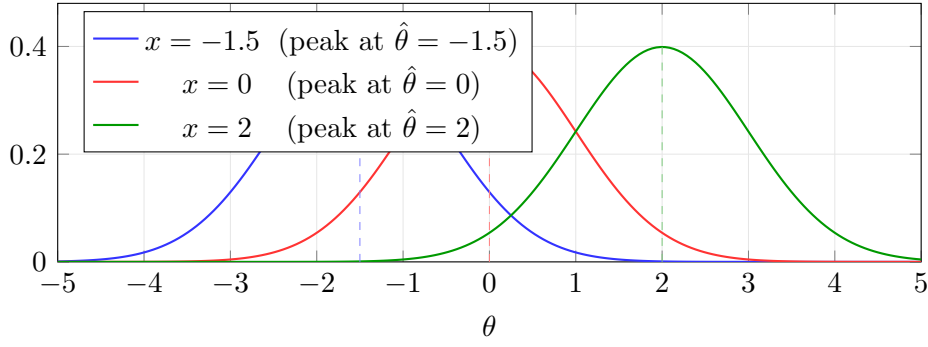
Key difference from the density. The density $f_\theta(x)$ integrates to 1 over x for every fixed θ (this is Property 1 of h). The pre-likelihood $L_x(\theta)$ need *not* integrate to 1 over θ ; it is not a density in θ . In the Bayesian framework one multiplies L_x by a prior $\pi(\theta)$ to obtain a proper posterior density, but frequentist theory makes no such requirement.

Graphical reading of $L_x(\theta)$. For $\mathcal{N}(\theta, \sigma^2)$ the pre-likelihood at the *number* x is

$$L_x(\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \theta)^2}{2\sigma^2}\right).$$

As a function of θ , this is bell-shaped, symmetric around $\theta = x$, and maximised at $\hat{\theta}(x) = x$. The plot below draws three such pre-likelihood curves at three different *numeric* values of x . When we later swap a number x for the random $X(\omega)$, each ω picks out one such curve (determined by where on the axis $X(\omega)$ happens to land).

Pre-likelihood $L_x(\theta) = f_\theta(x)$ for $\mathcal{N}(\theta, 1)$, three values of x



Each curve peaks at the corresponding x (since the MLE for $\mathcal{N}(\theta, 1)$ is $\hat{\theta}(x) = x$), but the shape — the curvature, the width — is the same. The curvature of $L_x(\theta)$ at the MLE is $-1/\sigma^2$, which is the (negative) Fisher information.

What the derivative of L_{x_1} looks like. A small foreshadowing of Chapter 3: if we differentiate the pre-likelihood $L_{x_1}(\theta)$ in θ , we get an odd S-shape. Concretely, for $\mathcal{N}(\theta, 1)$ at $x_1 = 0$,

$$\frac{dL_{x_1}}{d\theta} = \frac{d}{d\theta} \varphi(x_1 - \theta) = (x_1 - \theta) \varphi(x_1 - \theta) = -\theta \varphi(-\theta),$$

an odd function of θ around x_1 , with extrema at $\theta = x_1 \pm 1$. The Probabilist's hand-drawn picture below shows the shape.

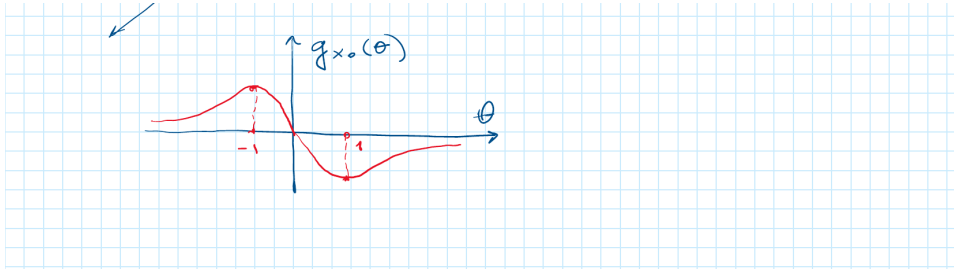


Figure 6: The derivative of the pre-likelihood, $\partial_\theta L_{x_1}(\theta)$, for $\mathcal{N}(\theta, 1)$ at $x_1 = 0$. The Probabilist's label $L_{x_1}(\theta)$ on the figure refers to the slice itself; what is drawn is its θ -derivative. The function is odd around $\theta = x_1$ with extrema at $\theta = x_1 \pm 1$, crosses zero at the MLE $\hat{\theta} = x_1$, and integrates to zero over θ (by symmetry, just as the pre-likelihood does). This curve will reappear in Chapter 3 as a θ -slice of the pre-score $s(x, \theta)$ at fixed $x = x_1$.

Not invariant under reparameterisation. The pre-likelihood is a function of θ , so its shape depends on how we parameterise the family. If $\phi = q(\theta)$ is a smooth bijection, the family $\{f_\theta\}$ can equally well be written as $\{\tilde{f}_\phi\}$ with $\tilde{f}_\phi := f_{q^{-1}(\phi)}$, and the pre-likelihood in the new parameter is

$$\tilde{L}_x(\phi) := \tilde{f}_\phi(x) = f_{q^{-1}(\phi)}(x) = L_x(q^{-1}(\phi)).$$

The *values* are preserved at corresponding points: the height of $\tilde{L}_x$ at ϕ equals the height of L_x at $\theta = q^{-1}(\phi)$. In particular the MLE is preserved as a point: $\hat{\phi}(x) = q(\hat{\theta}(x))$. But the *shape* — concavity, curvature, width — is generally *not* preserved unless q is affine. Reparameterising stretches and bends the horizontal axis non-uniformly, so a strictly concave L_x may become non-concave in ϕ , and the curvature at the MLE changes by the factor $(q'(\hat{\theta}))^{-2}$.

A small concrete example: for $\mathcal{N}(\theta, 1)$ the pre-likelihood $L_x(\theta)$ is a Gaussian bell in θ (strictly log-concave). Reparameterise by $\phi = \theta^3$ (smooth bijection). Then $\tilde{L}_x(\phi) = L_x(\phi^{1/3})$, which is no longer log-concave in ϕ near the origin: the cube-root rescaling flattens the curve on one side of zero and steepens it on the other. Same family, same data, same inferential content; different shape.

This caveat is structural — it has nothing to do with logs. We flag it now so that it does not have to be re-derived for the log-likelihood in Chapter 2.

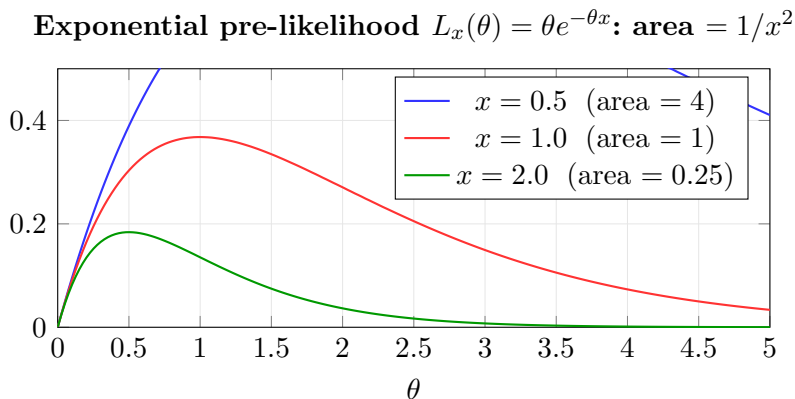
From pre-likelihood to likelihood: randomising the index. Each curve in the plot above corresponds to a fixed number x . The (proper) likelihood arises when, instead of choosing a particular x , we let nature choose: $\omega \in \Omega$ is drawn, and we look at the curve $L_{X(\omega)}$. The likelihood is therefore a *random curve*. Quantities computed from the likelihood (the score, its variance, the Fisher information) are then naturally random variables or expectations over ω .

The pre-likelihood does not integrate to 1 over θ . For the location-normal family the symmetry of the bell happens to make $\int L_x(\theta) d\theta = 1$, but this is a coincidence: the integral is a density in θ *only because* the density is a density in x and the two coordinates are coupled through $x - \theta$ alone.

A cleaner illustration uses the exponential model. With $f_\theta(x) = \theta e^{-\theta x}$ on $x > 0$, $\theta > 0$, the pre-likelihood at the fixed number x is $L_x(\theta) = \theta e^{-\theta x}$ on $\theta > 0$. Its integral over θ is

$$\int_0^\infty L_x(\theta) d\theta = \int_0^\infty \theta e^{-\theta x} d\theta = \frac{1}{x^2},$$

which equals 1 only at $x = 1$ and depends on x for every other observation. Normalisation of L_x in θ is therefore not a structural property; in this family it generically fails.



The three curves all have peak at $\hat{\theta}(x) = 1/x$ but different total areas: the pre-likelihood “stretches” or “shrinks” along θ depending on the observed x , and the area under the curve generally is not 1.

2 The Log-Likelihood

Before we differentiate the likelihood to obtain the score, it is worth pausing on the log transform itself. Every property of the log-likelihood that we will need — additivity for i.i.d. samples, preservation of the maximiser, concavity for exponential families — is a property of the function $\log L$ as a function of θ , *before* any derivatives are taken. Treating these properties separately, in their own chapter, prevents them from being silently absorbed into the score derivation.

2.1 Why the logarithm

The (pre-)likelihood is a product of densities and is typically very small in absolute terms; it can underflow even modest sample sizes. The log transform fixes several practical and structural inconveniences at once.

Numeric reason: products become sums. For an i.i.d. sample $x_1, \dots, x_n$ with density f_θ , the joint pre-likelihood at the numeric tuple $\mathbf{x}$ is

$$g_{\mathbf{x}}(\theta) = \prod_{i=1}^n f_\theta(x_i).$$

A product of n numbers, each of order 10^{-1} to 10^{-3} , underflows quickly. The log:

$$\ell_{\mathbf{x}}(\theta) := \log g_{\mathbf{x}}(\theta) = \sum_{i=1}^n \log f_\theta(x_i).$$

A sum of well-behaved numbers, no underflow.

Structural reason: additivity. The log transform turns the product structure of i.i.d. likelihoods into an additive one. This is more than a numerical convenience: it means that asymptotic theory (CLT, LLN) applies to the log-likelihood directly — to its first derivative the score $\sum \partial_\theta \log f_\theta(x_i)$, and to its second derivative the observed information $-\sum \partial_\theta^2 \log f_\theta(x_i)$. Without the log we would be working with products, where CLT-like results require multiplicative analogues that are far less clean.

Geometric reason: products of bell curves are still bells. For the location-normal family, $f_\theta(x) = (2\pi)^{-1/2} \exp(-(x - \theta)^2/2)$. The pre-likelihood is exp of a quadratic, so the log-likelihood is itself a quadratic in θ :

$$\ell_x(\theta) = -\frac{1}{2} \log(2\pi) - \frac{1}{2}(x - \theta)^2.$$

A parabola, opening downward, with vertex at $\theta = x$. We will see in a moment that downward-parabolic log-likelihoods are characteristic of a broad class (exponential families with quadratic sufficient statistic), not just the normal.

2.2 Definition and basic identities

Pre-log-likelihood (numeric index). For a fixed number $x \in \mathbb{R}$, let

$$\Theta_x := \{\theta \in \Theta : f_\theta(x) > 0\}$$

be the set of parameter values for which x has positive density. Define

$$\ell_x : \Theta_x \rightarrow \mathbb{R}, \quad \ell_x(\theta) := \log L_x(\theta) = \log f_\theta(x).$$

This is a deliberate departure from the textbook convention of extending $\log f_\theta(x)$ to all of Θ by setting it equal to $-\infty$ on the zero-density set. Per the Probabilist's recommendation, we restrict the domain instead: ℓ_x lives on the positivity set Θ_x , takes real values only, and the symbol $-\infty$ never enters our formulas.

The motivation is integration. Quantities like $\mathbb{E}_\theta[\ell_X]$ and $\mathbb{E}_\theta[(\partial_\theta \ell_X)^2]$ (the Fisher information) are integrals of log-of-density against the density itself. When log is allowed to take the value $-\infty$, every such integral inherits a $0 \cdot \infty$ ambiguity that has to be resolved by hand, conventions, or

measure-theoretic gymnastics ($0 \cdot \infty := 0$ being the standard but easily-forgotten patch). Restricting the domain to Θ_x from the outset eliminates the ambiguity: we never write $\log 0$, and every integral is over a region where the integrand is finite and the density is positive.

Remark 3 (When Θ_x depends on x). For families with parameter-independent support (Gaussian, Poisson, exponential families in canonical form), $\Theta_x = \Theta$ for every x in the support, and the restriction is vacuous. The restriction bites for families with θ -dependent support such as $\text{Uniform}[0, \theta]$: there, $\Theta_x = (x, \infty)$ for $x > 0$, and pre-log-likelihood is defined only for θ strictly larger than the observed value. This is precisely the family where the regularity conditions for the score identity fail (Remark in Section 3.3); restricting the domain makes the failure transparent rather than swept under a $-\infty$ rug.

The i.i.d. case and why we restrict to it. For an i.i.d. tuple $\mathbf{x} = (x_1, \dots, x_n)$ the joint positivity set is the intersection,

$$\Theta_{\mathbf{x}} = \bigcap_{i=1}^n \Theta_{x_i},$$

and the joint pre-log-likelihood factors additively:

$$\ell_{\mathbf{x}}(\theta) = \sum_{i=1}^n \ell_{x_i}(\theta), \quad \theta \in \Theta_{\mathbf{x}}.$$

Both expressions rely heavily on the i.i.d. assumption. Two things are happening, and it is worth pulling them apart.

First, the joint density factorises:

$$f_{\theta}^{(n)}(x_1, \dots, x_n) = \prod_{i=1}^n f_{\theta}(x_i).$$

This is why the log turns into a sum.

Second, and more fundamentally, a single scalar parameter θ suffices to describe the distribution of the entire random vector $\mathbf{X} = (X_1, \dots, X_n)$. In the i.i.d. setting, θ parameterises the common marginal $X_i \sim f_{\theta}$, and the joint distribution is fully determined by θ together with the independence assumption. No additional parameters are needed to specify the dependence structure — there isn't any.

If we drop i.i.d., both of these break. Suppose $(X_1, \dots, X_n)$ has an arbitrary joint distribution. Then a single θ governing the marginal of each X_i is no longer enough to characterise the joint distribution of $\mathbf{X}$: we also need to specify the dependence between the X_i 's. In general, we need a richer parameter $\boldsymbol{\eta}$ (typically vector-valued, sometimes infinite-dimensional) indexing a family of joint densities $\{f_{\boldsymbol{\eta}}^{(n)}\}$ on $\mathbb{R}^n$, and the pre-log-likelihood becomes

$$\ell_{\mathbf{x}}(\boldsymbol{\eta}) = \log f_{\boldsymbol{\eta}}^{(n)}(x_1, \dots, x_n),$$

which does *not* decompose into a sum in any useful way. Two representative non-i.i.d. cases give a sense of where the difficulty lies:

- **Identically distributed but dependent** (e.g. a stationary time series $\{X_t\}$). Each X_t has the same marginal f_{θ} , but the joint density depends on θ *and* on autocorrelation/copula parameters. The marginal θ alone is not a sufficient parameterisation of $\mathbf{X}$.
- **Independent but not identically distributed** (e.g. regression with non-random X , $Y_i \sim \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$). The joint density factorises, $f^{(n)} = \prod_i f_{\theta_i}$, but each marginal has its own parameter $\theta_i = \beta_0 + \beta_1 x_i$. The right object is the vector $\boldsymbol{\eta} = (\beta_0, \beta_1, \sigma^2)$, not a single θ .

In both cases the log-likelihood is well-defined, but θ alone is the wrong index. Working out the score and Fisher information for non-i.i.d. models requires care: the additive structure of the i.i.d. score breaks, asymptotic theory needs CLTs adapted to the dependence (mixing conditions, martingale CLT) or to the heterogeneity (Lindeberg conditions), and the information matrix may fail to be block-diagonal in ways that complicate inversion.

We do not pursue the non-i.i.d. direction in this text. From here on, the sample is i.i.d. unless stated otherwise, and θ is a scalar parameter of the common marginal. The flagging above is meant only to say: this is a real boundary, and the clean identities of the next two chapters live on the i.i.d. side of it.

Log-likelihood (random index). Substituting $X_i(\omega)$ for the numeric tuple gives the **log-likelihood**:

$$\mathcal{L}(\omega, \theta) := \log L(\omega, \theta) = \sum_{i=1}^n \log f_{\theta}(X_i(\omega)), \quad \theta \in \Theta_{X(\omega)},$$

where $\Theta_{X(\omega)} = \bigcap_i \Theta_{X_i(\omega)}$ is now itself random (a random subset of Θ). This is a random function of θ . Each ω produces one realised log-likelihood curve, defined on its own positivity slice.

Monotone equivalence with the likelihood. Since log is strictly increasing on $(0, \infty)$, the maximiser of ℓ_x over Θ_x equals the maximiser of L_x over the same set:

$$\arg \max_{\theta \in \Theta_x} L_x(\theta) = \arg \max_{\theta \in \Theta_x} \ell_x(\theta).$$

And since $L_x(\theta) = 0$ off Θ_x , restricting the argmax to Θ_x loses no maxima: any θ with $L_x(\theta) = 0$ is dominated by points where $L_x > 0$. So MLE theory is universally formulated in terms of the log-likelihood: nothing is lost by passing to logs, and additivity is gained.

2.3 Concavity and exponential families

Quadratic log-likelihood for the location-normal family. We saw above that for $\mathcal{N}(\theta, 1)$, $\ell_x(\theta) = -\frac{1}{2}(x - \theta)^2 + \text{const}$ — a downward parabola. For an i.i.d. sample this becomes

$$\ell_{\mathbf{x}}(\theta) = -\frac{1}{2} \sum_{i=1}^n (x_i - \theta)^2 + \text{const},$$

still a downward parabola in θ , with vertex at the sample mean $\bar{x} = \frac{1}{n} \sum x_i$. Concavity is uniform in $\mathbf{x}$, and the maximiser is unique.

General exponential families: concavity in the natural parameter. A one-parameter exponential family in canonical form has density

$$f_{\theta}(x) = h_0(x) \exp(\theta T(x) - A(\theta)),$$

where T is the sufficient statistic and $A(\theta)$ is the log-partition function. Then

$$\ell_x(\theta) = \log h_0(x) + \theta T(x) - A(\theta).$$

A fundamental result of exponential-family theory is that A is *convex*; therefore ℓ_x is *concave* in θ (strictly, when the family is minimal and identifiable). Concavity of the log-likelihood implies that any stationary point is a global maximum, and gradient-based optimisation is well-posed. This is the deep reason that MLE in exponential families is computationally tame.

Non-concavity outside exponential families. Concavity is a property of exponential families, not of likelihoods in general. Mixture models, hidden-state models, and many non-identifiable models have multimodal log-likelihoods. This is where EM and other iterative schemes earn their keep.

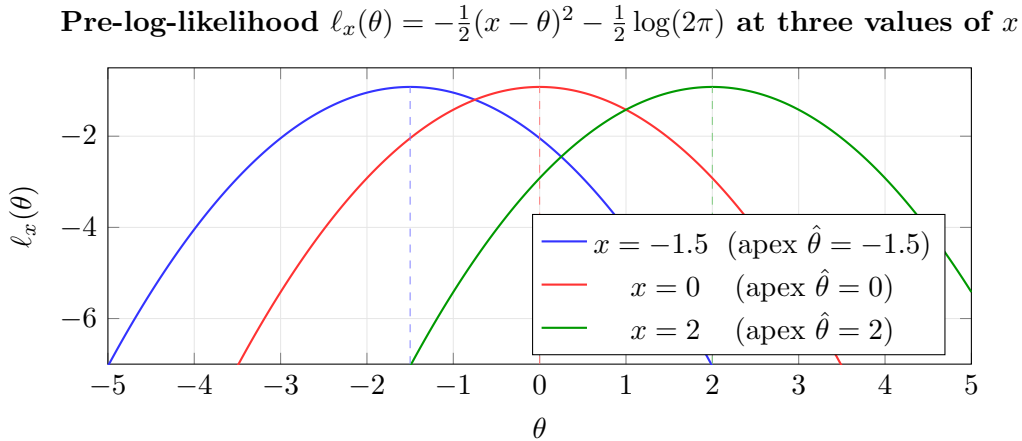
2.4 The log-likelihood surface and its slices

In direct analogy with $h(x, \theta) = f_\theta(x)$, define

$$\tilde{h}(x, \theta) := \log f_\theta(x), \quad (x, \theta) \in D, \quad D := \{(x, \theta) : f_\theta(x) > 0\}.$$

The domain $D \subset \mathbb{R} \times \Theta$ is the positivity set of the family — the same region on which all our log-likelihood slices live. Outside D , $\tilde{h}$ is undefined; we do not assign it the value $-\infty$. This is the surface that the log-likelihood and pre-log-likelihood slice up. The two relevant directions:

- **Fixed θ , varying x :** $\tilde{h}(\cdot, \theta_0)$ on the θ_0 -slice of D is the log-density. It does *not* integrate to 1 over x ; instead, $\int e^{\tilde{h}} dx = 1$ (the integral being over the positivity slice, where the integrand f_{θ_0} is automatically zero off-slice anyway). Differential entropy is $-\mathbb{E}_{\theta_0}[\tilde{h}(X, \theta_0)] = -\int_{\{f_{\theta_0} > 0\}} f_{\theta_0} \log f_{\theta_0} dx$.
- **Fixed x , varying θ :** $\tilde{h}(x_1, \cdot)|_{\Theta_{x_1}} = \ell_{x_1}$ is the pre-log-likelihood, the workhorse of all that follows.



Compare to the pre-likelihood plot of Section 1.4: same peak locations, but the curves are now strict parabolas opening downward. The information about where the apex sits is preserved; what changes is the shape further from the apex — parabolic decay (log-scale) vs. Gaussian decay (linear scale).

2.5 What the log-likelihood does not tell us

A few honest caveats before we differentiate. The log-likelihood is the central object of frequentist inference, but it has its limits:

- **Not invariant under reparameterisation.** Like the (pre-)likelihood itself — see the discussion in Section 1.4 — the log-likelihood is not invariant under smooth reparameterisations $\phi = q(\theta)$: values match at corresponding points, but shape (concavity, curvature) is generally not preserved unless q is affine. Taking the log does not save us here.

- **Not directly a posterior log-density.** In Bayesian inference one adds $\log \pi(\theta)$ (the log prior) to obtain the log posterior, up to a normalising constant. The log-likelihood by itself is not interpretable as a log-density in θ .
- **Comparable up to additive constants.** $\ell_x(\theta) + C$ contains the same inferential information as $\ell_x(\theta)$; only relative log-likelihoods matter. This is why log-likelihood plots are often shown with the maximum subtracted, plotting $\ell_x(\theta) - \ell_x(\hat{\theta})$.

We now have the structural setup to take derivatives. In the next chapter, $\partial_\theta \ell$ is the score, and $-\partial_\theta^2 \ell$ the observed information — objects *built from* the log-likelihood, not replacements for it.

3 Score, Fisher Information, and the Information Identity

We now differentiate ℓ with respect to θ . The first derivative is the score; the negative second derivative (in expectation) is the Fisher information. Before listing the score's other properties, we cash in on the work done in Chapter 7: Property 3 (the vanishing θ -integral of $\partial_\theta h$) was waiting to deliver the zero-mean identity of the score, and now we collect.

3.1 The density surface and what is differentiated

We will work with the same surface $h(x, \theta) = f_\theta(x)$ — not because it is new, but because it is the right substrate. The score is the θ -derivative of the log of this surface, restricted to the data slice:

$$s(\theta) := \partial_\theta \ell_x(\theta) = \frac{\partial_\theta h(x, \theta)}{h(x, \theta)} = \partial_\theta \log f_\theta(x).$$

The score is a function of θ indexed by x (pre-score) or by ω (score proper, after substituting $X(\omega)$).

3.2 The semi-elasticity operator: algebra before semantics

The score is one instance of a single differential operator applied to a positive function. Before specialising to likelihoods, it is worth collecting the operator's general properties — they explain several features of the score (additivity for i.i.d. samples, behaviour under reparameterisation, Fisher information identity) at the algebraic level, without any probability.

Definition. Let $\mathcal{P}$ be the set of positive, differentiable real-valued functions on an open set $U \subset \mathbb{R}$. The **semi-elasticity operator** is the map

$$D : \mathcal{P} \longrightarrow F(U), \quad D[f](t) := \frac{f'(t)}{f(t)} = (\log f)'(t),$$

where $F(U)$ is just “the set of real-valued functions on U ” — *not* necessarily continuous ones. The point is subtle and is worth being explicit about.

Remark 4 (Why we cannot write $D : \mathcal{P} \rightarrow C(U)$). A function f being differentiable on U does not imply that its derivative f' is continuous on U . The classical counterexample is

$$f(t) = \begin{cases} t^2 \sin(1/t), & t \neq 0, \\ 0, & t = 0, \end{cases}$$

defined on all of $\mathbb{R}$. This f is differentiable everywhere (one checks $f'(0) = 0$ directly from the limit definition), but

$$f'(t) = 2t \sin(1/t) - \cos(1/t) \quad (t \neq 0),$$

which oscillates between roughly -1 and $+1$ as $t \rightarrow 0$ and has no limit at 0 . So $f \in \mathcal{P}$ (after shifting upward to keep it positive) does not imply $D[f] \in C(U)$.

Such examples are pathological — they are constructed on purpose and do not arise from any parametric family we care about — but they show that the codomain “ $C(U)$ ” I wrote earlier is technically wrong. If we want a clean operator with continuous output, we have to restrict the source: let $\mathcal{P}_1 \subset \mathcal{P}$ be the set of positive C^1 functions on U . Then

$$D : \mathcal{P}_1 \longrightarrow C(U)$$

is genuinely continuous-valued, because $f \in \mathcal{P}_1$ has $f' \in C(U)$, $1/f \in C(U)$ (since $f > 0$ continuous), and the product f'/f is then in $C(U)$.

For everything we do with likelihoods of standard parametric families, the underlying densities are C^∞ in θ , so the pre-log-likelihoods live in $\mathcal{P}_1$ (in fact in C^∞), and the codomain question never bites. We retain the wider domain $\mathcal{P}$ for generality but acknowledge that $D[f]$ is then a “mere” function, not automatically a continuous one.

Two reformulations:

$$D[f] = \frac{f'}{f} = (\log f)'.$$

The first is the algebraic ratio of f' and f ; the second is the derivative of $\log f$. These are the two genuinely distinct ways of writing D ; the choice of notation for the derivative (f' vs. df/dt vs. $\partial_t f$) is immaterial. Because f has just one (scalar) argument t , the appropriate symbol is the *total* derivative d/dt ; we use f' throughout for brevity, reserving ∂_t for the partial derivative in θ later, once the second argument x enters the picture. The semi-elasticity name comes from definition (iii) of Section 3.4.

D as a composition: two paths through intermediate spaces. The two reformulations above correspond to two different *factorisations* of D through more elementary operators. Set up the relevant function spaces first:

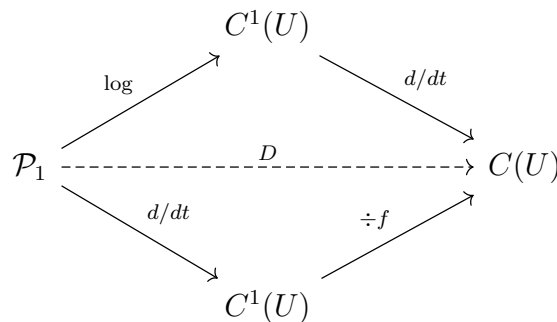
$$\begin{aligned} \mathcal{P}_1 &:= \{f : U \rightarrow \mathbb{R} : f > 0, f \in C^1\} && \text{(positive, } C^1\text{)} \\ C(U) &:= \text{continuous real-valued functions on } U \\ C^1(U) &:= \text{continuously differentiable functions on } U \end{aligned}$$

(We restrict to $\mathcal{P}_1$ in this diagram so the codomain of D is genuinely $C(U)$; see Remark 4 for what happens on the wider class $\mathcal{P}$.)

The three elementary operators we will compose are:

$$\begin{array}{lll} \log : \mathcal{P}_1 \rightarrow C^1(U), & f \mapsto \log f & \text{(pointwise: zeroth-order local)} \\ d/dt : C^1(U) \rightarrow C(U), & g \mapsto g' & \text{(neighbourhood: first-order local)} \\ \div : C^1(U) \rightarrow C(U), & f' \mapsto f'/f & \text{(pointwise, with } f \in \mathcal{P}_1 \text{ fixed)} \end{array}$$

Both paths from $\mathcal{P}_1$ to $C(U)$ give the same answer:



Path A (upper): $f \xrightarrow{\log} \log f \xrightarrow{d/dt} (\log f)'$. First take the log pointwise; then differentiate.

Path B (lower): $f \xrightarrow{d/dt} f' \xrightarrow{\div f} f'/f$. First differentiate; then divide pointwise by f (which we still have available, since we are tracking f alongside the computation of f').

Equality of the two paths is the chain rule applied to $\log$:

$$(\log f)'(t) = \frac{1}{f(t)} \cdot f'(t) = \frac{f'(t)}{f(t)}.$$

The two paths are equal as operators, but they have different rhetorical force. Path A says “differentiate the log-likelihood” — the standard textbook framing. Path B says “divide the gradient by the value” — the framing that makes clear why constants kill ($D[c \cdot f] = D[f]$) and why the score respects multiplication. In computations we will freely move between them.

Intermediate spaces have different “geographies”. A subtle point worth flagging: the two intermediate spaces in the diagram are both copies of $C^1(U)$, but they are reached by quite different maps from $\mathcal{P}_1$.

Path A takes the log first, going from $\mathcal{P}_1$ (positive functions) into a copy of $C^1(U)$ that may contain unbounded negative values (since $\log f \rightarrow -\infty$ as $f \rightarrow 0^+$). The positivity of f is no longer visible; only the absence of zero values is preserved (as $\log f$ being finite).

Path B differentiates first, going into a copy of $C^1(U)$ that can contain negative values too (since f' can be negative wherever f is decreasing). The positivity of f is preserved, but only implicitly, as the fact that we are still allowed to divide by it in the next step.

The two intermediate routes pass through different “regions” of function space, but converge to the same answer.

The two paths have different domains. A sharper version of the previous point: viewed as abstract operators on a class of functions, Paths A and B are not defined on the same class.

Path B (d/dt then $\div f$) is defined wherever f is *differentiable and non-vanishing*. The output f'/f is a formal semi-elasticity, available even for functions that take negative values (e.g. $f(t) = -e^{-t}$ gives $f'/f = -1$).

Path A ($\log$ then d/dt) is defined only where f is *differentiable and strictly positive*, since $\log$ requires positivity. On a function that takes negative values, Path A fails at the first step.

So Path B has a wider natural domain than Path A. On the intersection — positive differentiable f — the two paths agree and the diagram of the previous paragraph commutes. But the “semi-elasticity” f'/f as a bare ratio exists for a larger class of f than the log-derivative interpretation does.

For likelihood theory this distinction is moot: densities are non-negative, and on their positivity set they are strictly positive, so both paths apply. But when the same operator appears outside the likelihood context — for instance in growth-rate calculations in economics, where the underlying quantity may be negative — only Path B remains a meaningful definition.

Locality. A key feature of D that we will use without comment is that it is a *local* operator: $D[f](t_0)$ depends only on f in an arbitrarily small neighbourhood of t_0 . More precisely, each constituent step of Paths A and B is local in a sharp sense:

- **Pointwise (zeroth-order locality).** The map $f \mapsto \log f$ is even more local than “local”: $\log f(t_0)$ depends only on f at the single point t_0 , not on a neighbourhood. It is a pointwise transformation. The same is true of division by f , multiplication by a function, and any composition of f with a fixed real-valued function g , giving $g \circ f$.

- **Neighbourhood (first-order locality).** Differentiation $f \mapsto f'$ requires more than a single value: $f'(t_0)$ is defined as a limit of difference quotients $(f(t_0 + h) - f(t_0))/h$, so we need f on a small interval containing t_0 . But the interval can be arbitrarily small. Second-order locality means: any modification of f *outside* a neighbourhood of t_0 , however large, leaves $f'(t_0)$ unchanged.

The composite $D = (d/dt) \circ \log$ inherits the weaker of the two: $D[f](t_0)$ requires f on a neighbourhood (first-order local).

This contrasts sharply with the *integral* operators that will appear when we take expectations. The map

$$f \mapsto \mathbb{E}_\theta[f] = \int f(x) p(x | \theta) dx$$

is *global*: its value at θ depends on f over the entire data axis. Changing f on any region of positive measure changes the integral. This is why differentiating before integrating (Property 3 of Chapter 1: $\int \partial_\theta h dx = 0$) is non-trivial: it interchanges a local operator ∂_θ with a global operator $\int dx$, and the interchange needs the regularity conditions we keep citing. The score identity $\mathbb{E}_\theta[s] = 0$ is, structurally, an identity relating the action of one local operator (D) and one global operator ($\mathbb{E}_\theta$) on the same surface h .

Domain. $D[f]$ is well-defined wherever $f > 0$ and f is differentiable. Positivity matters: $\log f$ has to make sense, and the ratio f'/f fails to be continuous (or even defined) where f vanishes. This is the operator-level reason for restricting the domain to the positivity set Θ_x when defining the pre-log-likelihood (Section 1.4): outside Θ_x , D is not applicable.

For non-smooth f (e.g. Laplace density, which has a kink at μ), $D[f]$ is undefined at the kink and one needs subgradients or distributional derivatives; we set such cases aside.

Property 1: Not additive — multiplicative. The operator D is *not* a linear / additive operator on $\mathcal{P}$. Adding two positive functions in general gives a new positive function, but

$$D[f + g] = \frac{f' + g'}{f + g} \neq D[f] + D[g]$$

in general. No clean formula for the sum exists. This is the reason we never write a score for “the likelihood plus a constant” or for mixture models in closed form.

What D *does* respect is **multiplication**:

$$D[fg] = \frac{(fg)'}{fg} = \frac{f'g + fg'}{fg} = \frac{f'}{f} + \frac{g'}{g} = D[f] + D[g].$$

Multiplication of f ’s becomes addition of $D[f]$ ’s. Equivalently:

$$D[fg] = D[f] + D[g], \quad D[f/g] = D[f] - D[g], \quad D[f^\alpha] = \alpha D[f] \quad (\alpha \in \mathbb{R}).$$

This is exactly why i.i.d. samples give additive scores: the joint likelihood $L(\theta) = \prod_i f_\theta(x_i)$ is a product, and $D = \partial_\theta \log$ converts that product into a sum $s(\theta) = \sum_i \partial_\theta \log f_\theta(x_i)$. The additivity of the score is not an accident — it is the operator algebra of D applied to a product.

Property 2: Iteration. Apply D twice (with respect to the same variable):

$$D^2[f] = D\left[\frac{f'}{f}\right] = \frac{d}{dt}\left(\frac{f'}{f}\right) = \frac{f''}{f} - \left(\frac{f'}{f}\right)^2 = \frac{f''}{f} - D[f]^2.$$

So $D^2[f]$ does *not* equal f''/f ; there is a quadratic correction. Rearranging,

$$\frac{f''}{f} = D^2[f] + D[f]^2 = (\log f)'' + ((\log f)')^2.$$

This identity will reappear shortly as the **information identity**: when f is a likelihood, $-\mathbb{E}[D^2\ell]$ (the expected observed information) equals $\mathbb{E}[D[\ell]^2]$ (the variance of the score), and the cross term f''/f has zero expectation under the true parameter — a direct consequence of $\int \partial_\theta^2 h dx = 0$, which is Property 3 differentiated again.

Property 3: Reparameterisation (chain rule). If $\phi = q(\theta)$ is a smooth bijection and $\tilde{f}(\phi) = f(q^{-1}(\phi))$, then by the chain rule

$$D_\phi[\tilde{f}](\phi) = \frac{\tilde{f}'(\phi)}{\tilde{f}(\phi)} = \frac{f'(q^{-1}(\phi)) \cdot (q^{-1})'(\phi)}{f(q^{-1}(\phi))} = (q^{-1})'(\phi) \cdot D_\theta[f](q^{-1}(\phi)).$$

The semi-elasticity transforms by the *Jacobian* of the inverse reparameterisation, not invariantly. In particular, the score is *not* invariant under reparameterisation: $s_\phi(\phi) = s_\theta(\theta)/q'(\theta)$. This is the operator-level statement of the caveat from Section 1.4.

Property 4: Zero of $D[f]$ = stationary point of $\log f$ = stationary point of f . Since $D[f] = (\log f)'$ and $\log$ is strictly increasing, $D[f](t_0) = 0 \iff (\log f)'(t_0) = 0 \iff f'(t_0) = 0$. Zeros of the semi-elasticity coincide with stationary points of the original f . Applied to the likelihood, this is the first-order condition for the MLE: $s(\hat{\theta}) = 0 \iff L'(\hat{\theta}) = 0$.

Property 5: Non-injectivity and the fibres of D . D is not injective: many different f 's give the same $D[f]$. This is worth comparing to the ordinary derivative, since the fibres look different in the two cases.

Fibres of d/dt . If $g'_1 = g'_2$, then $g_1 - g_2$ has zero derivative, so $g_1 - g_2 = c$ for some constant c . The pre-image of any given derivative is a family of *parallel translates*:

$$(d/dt)^{-1}(g') = \{g + c : c \in \mathbb{R}\}.$$

Vertical shifts are invisible to differentiation.

Fibres of D . If $D[f_1] = D[f_2]$, then $(\log f_1)' = (\log f_2)'$, so $\log f_1 - \log f_2 = \text{const}$, i.e. $f_1/f_2 = \text{const}$. The pre-image is a family of *positive proportions*:

$$D^{-1}(D[f]) = \{c \cdot f : c > 0\}.$$

Multiplicative rescalings are invisible to D .

The parallel is clean: differentiation forgets additive constants, the semi-elasticity operator forgets multiplicative ones. This is exactly the operator-level statement that likelihoods are defined only up to a positive multiplicative constant — the score (and everything built from it: Fisher information, MLE, tests) depends on the equivalence class $\{c \cdot L : c > 0\}$, not on the particular representative. Two likelihoods that differ by a normalising constant give the same inference.

In compact form: D factors through the quotient $\mathcal{P}_1/\sim$, where $f \sim g \iff f = cg$ for some $c > 0$.

Summary table.

Property of D	Consequence for the score
$D[fg] = D[f] + D[g]$	i.i.d. scores are additive: $s = \sum_i s_i$
$D^2[f] = f''/f - D[f]^2$	information identity (after expectation)
$D_\phi[\tilde{f}] = (q^{-1})' \cdot D_\theta[f]$	score not invariant under reparam.
$D[f](t_0) = 0 \iff f'(t_0) = 0$	score MLE condition = likelihood MLE
$D^{-1}(D[f]) = \{cf : c > 0\}$	score depends on shape, not scale
not additive: $D[f + g] \neq D[f] + D[g]$	mixtures have no closed-form score

The next subsection applies D to the density to recover the zero-mean property of the score.

3.3 From Property 3 to the score identity

Property 3 of Chapter 1 says $\int \partial_\theta h \, dx = 0$. Now divide and multiply by h inside the integral (assuming $h > 0$):

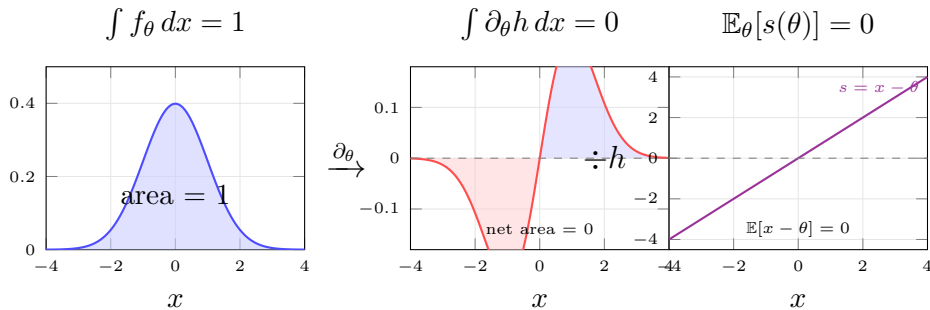
$$0 = \int_{\mathbb{R}} \frac{\partial h}{\partial \theta} \, dx = \int_{\mathbb{R}} \frac{\partial h / \partial \theta}{h(x, \theta)} \cdot h(x, \theta) \, dx = \int_{\mathbb{R}} \frac{\partial \log h}{\partial \theta} \cdot h(x, \theta) \, dx = \mathbb{E}_\theta \left[\frac{\partial \log f_\theta}{\partial \theta} \right] = \mathbb{E}_\theta[s(\theta)].$$

So the score has zero mean *under the true parameter* — not by coincidence, not as a separate axiom, but as a direct consequence of the fact that the density integrates to one. The derivation is a chain:

$$\underbrace{\int h \, dx = 1}_{\text{density axiom}} \xrightarrow{\text{diff. w.r.t. } \theta} \underbrace{\int \partial_\theta h \, dx = 0}_{\text{Property 3}} \xrightarrow{\div h, \times h} \underbrace{\mathbb{E}_\theta[s] = 0}_{\text{score identity}}.$$

Remark 5 (What the regularity condition does). The step “differentiate under the integral” requires justification — one needs to exchange $\frac{d}{d\theta}$ and $\int$. Sufficient conditions (dominated convergence) are satisfied for exponential families, Gaussian models, and most textbook parametric models. For the Uniform $[0, \theta]$ family this interchange fails (the support depends on θ), and indeed $\mathbb{E}[s] \neq 0$. The score is $s = -n/\theta < 0$ everywhere, which is consistent with the failure of the zero-mean property.

Graphical summary of the chain. The figure below shows the three steps as a sequence of panels.



Definition 6. The **score** is the derivative of the log-likelihood with respect to the parameter:

$$s(\theta) = \frac{\partial}{\partial \theta} \ell(\theta) = \frac{\partial}{\partial \theta} \log p(\mathbf{x} \mid \theta).$$

For an i.i.d. sample: $s(\theta) = \sum_{i=1}^n \frac{\partial}{\partial \theta} \log p(x_i \mid \theta)$.

Two fundamental properties:

(i) **At the MLE, score is zero:** $s(\hat{\theta}_{\text{MLE}}) = 0$ — the first-order condition for an interior maximum (see Section 3.7 for exceptions).

(ii) **Under the true parameter, score has zero mean:** $\mathbb{E}_\theta[s(\theta)] = 0$. Proof: differentiate $\int p(x | \theta) dx = 1$ under the integral sign, use $\partial_\theta \log p = \frac{1}{p} \partial_\theta p$.

Several parameters: the gradient. When $\theta \in \mathbb{R}^p$, the score is the gradient vector $\mathbf{s}(\theta) = \nabla_\theta \ell(\theta)$; the MLE satisfies $\mathbf{s}(\hat{\theta}) = 0$.

3.4 Score as a semi-elasticity of the likelihood

There are several notions of elasticity; they differ in which argument is differentiated in absolute vs. relative (logarithmic) terms. Let $f(t) > 0$ be a differentiable function.

Three definitions. (i) **Full (log-log) elasticity** — relative change in f per relative change in t :

$$\varepsilon^{\log-\log}(t) = \frac{d \log f}{d \log t} = t \cdot \frac{d \log f}{dt} = \frac{t}{f(t)} \cdot f'(t).$$

Dimensionless. Both f and t enter through logarithms.

(ii) **Semi-elasticity of f with respect to $\log t$ (log-level)** — absolute change in f per relative change in t :

$$\varepsilon^{f-\log t}(t) = \frac{df}{d \log t} = t \cdot f'(t).$$

(iii) **Semi-elasticity of $\log f$ with respect to t (log-level, the other way)** — relative change in f per absolute change in t :

$$\varepsilon^{\log f-t}(t) = \frac{d \log f}{dt} = \frac{f'(t)}{f(t)}.$$

This takes the logarithm of f only, not of t .

The score is definition (iii). Applying it to the likelihood $L(\theta)$:

$$\frac{d \log L}{d \theta} = \frac{L'(\theta)}{L(\theta)} = s(\theta).$$

The score is the semi-elasticity of the likelihood L with respect to the parameter θ : the proportional change in the likelihood per unit absolute change in θ . It takes the logarithm of L but *not* of θ .

Remark 7 (Relation to the full elasticity). The full log-log elasticity of L is $\varepsilon^{\log-\log}(\theta) = \theta \cdot s(\theta)$. This is *not* the score; it differs by a factor of θ . At the MLE both vanish (since $s(\hat{\theta}) = 0$ and $\hat{\theta} \neq 0$ generically), so they give the same optimality condition, but they are different objects.

3.5 Fisher information and the information identity

For the normal model $\mathcal{N}(\mu, \sigma^2)$ the two Fisher informations are:

$$\mathcal{I}_\mu = \frac{1}{\sigma^2}, \quad \mathcal{I}_{\sigma^2} = \frac{1}{2\sigma^4}.$$

The information matrix (both parameters jointly) is diagonal:

$$\mathcal{I}(\mu, \sigma^2) = \begin{pmatrix} 1/\sigma^2 & 0 \\ 0 & 1/(2\sigma^4) \end{pmatrix}.$$

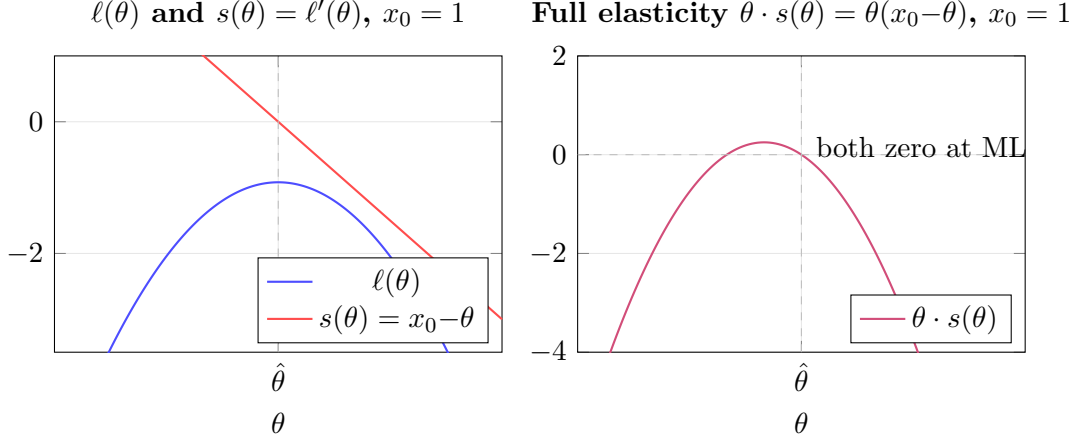


Figure 7: Left: log-likelihood $\ell(\theta) = \log L(\theta)$ and score $s(\theta) = \ell'(\theta) = d \log L / d\theta$ — the semi-elasticity (definition iii). Right: full log-log elasticity $\theta \cdot s(\theta) = \partial \log L / \partial \log \theta$ (definition i). Both vanish at the MLE; they are different functions of θ .

3.5.1 The three-way identity

The Fisher information satisfies the following chain of equalities:

$$\mathcal{I}(\theta) = \text{Var}_\theta[s(\theta)] = \mathbb{E}_\theta[s(\theta)^2] = -\mathbb{E}_\theta\left[\frac{\partial^2 \ell}{\partial \theta^2}\right].$$

The first equality holds because $\mathbb{E}[s(\theta)] = 0$ (under the true parameter), so variance equals the second moment. The third equality — $\mathbb{E}[s^2] = -\mathbb{E}[\ell'']$ — is the **information equality** (Bartlett's second identity).

3.5.2 Proof of the information equality (the Probabilist, $n = 1$)

We give the Probabilist's proof for $n = 1$, which is slightly more detailed in its intermediate steps than the streamlined version below.

Step 1: second derivative of a logarithm. Let $y = f(\theta)$ and $z = \ln y = \ln f(\theta)$. Then

$$\frac{dz}{d\theta} = \frac{d}{d\theta} \ln f(\theta) = \frac{1}{f(\theta)} \cdot f'(\theta),$$

and differentiating once more:

$$\frac{d^2 z}{d\theta^2} = \frac{d}{d\theta} \left(\frac{f'(\theta)}{f(\theta)} \right) = \frac{f''(\theta) \cdot f(\theta) - (f'(\theta))^2}{(f(\theta))^2} = \frac{f''(\theta)}{f(\theta)} - \left(\frac{f'(\theta)}{f(\theta)} \right)^2.$$

So for any positive differentiable f :

$$(\ln f)'' = \frac{f''}{f} - \left(\frac{f'}{f} \right)^2. \quad (*)$$

Step 2: introduce the second variable x . Abusing notation slightly (using f for what is usually written p or h), set $f(x, \theta)$ a joint function and define

$$z(x, \theta) := \frac{\partial^2}{\partial \theta^2} \ln f(x, \theta) = \frac{\partial^2 f / \partial \theta^2}{f(x, \theta)} - \left(\frac{\partial f / \partial \theta}{f(x, \theta)} \right)^2.$$

Step 3: substitute $x \mapsto X(\omega)$ and take expectation. For an abstract probability measure P on Ω :

$$\mathbb{E}_P[z(X(\omega), \theta)] = \mathbb{E}_P\left[\frac{\partial^2 f / \partial \theta^2(X(\omega), \theta)}{f(X(\omega), \theta)}\right] - \mathbb{E}_P\left[\left(\frac{\partial f / \partial \theta(X(\omega), \theta)}{f(X(\omega), \theta)}\right)^2\right].$$

Now suppose P has density $p(x)$ (with respect to Lebesgue measure). The expectations become integrals. The second term on the right is $\mathbb{E}_P[s^2]$ where $s(x, \theta) = \partial_\theta \ln f(x, \theta)$ is the score. We focus on the *minuend* (first term):

$$\text{minuend} = \mathbb{E}_P\left[\frac{\partial^2 f(x, \theta) / \partial \theta^2}{f(x, \theta)}\right] = \int_{\mathbb{R}} \frac{\partial^2 f(x, \theta) / \partial \theta^2}{f(x, \theta)} \cdot p(x) dx.$$

For a general p we cannot simplify this further.

Step 4: specialise to $p(x) = f(x, \theta)$. Under the true model the sampling density equals $f(\cdot, \theta)$, so $p(x) = f(x, \theta)$ and the minuend becomes

$$\text{minuend}|_{p=f} = \int_{\mathbb{R}} \frac{\partial^2 f(x, \theta) / \partial \theta^2}{f(x, \theta)} \cdot f(x, \theta) dx = \int_{\mathbb{R}} \frac{\partial^2 f(x, \theta)}{\partial \theta^2} dx.$$

Under the regularity assumption that differentiation and integration may be exchanged:

$$\text{minuend}|_{p=f} = \frac{\partial^2}{\partial \theta^2} \int_{\mathbb{R}} f(x, \theta) dx = \frac{\partial^2}{\partial \theta^2} 1 = \frac{\partial}{\partial \theta} 0 = 0.$$

Conclusion. With the minuend equal to zero, the identity (*) under $\mathbb{E}[\cdot]$ gives

$$\mathbb{E}[z(X, \theta)] = 0 - \mathbb{E}[s^2(X, \theta)],$$

i.e. $\mathbb{E}[z] = -\mathbb{E}[s^2]$. Since $z = (\ln f)'' = \ell''$ and $s = (\ln f)' = \ell'$, this reads

$$\boxed{\mathcal{I}_{f(\cdot, \theta)}(\theta) = \mathbb{E}[s^2(X, \theta)] = -\mathbb{E}\left[\frac{\partial^2}{\partial \theta^2} \ell(X, \theta)\right].}$$

3.5.3 Proof of the information equality (streamlined)

We know $\mathbb{E}_\theta[s(\theta)] = 0$ for all θ , i.e.

$$\int s(x, \theta) p(x | \theta) dx = 0 \quad \forall \theta.$$

Differentiate both sides with respect to θ :

$$\int \frac{\partial s}{\partial \theta} p dx + \int s \frac{\partial p}{\partial \theta} dx = 0.$$

The first integral is $\mathbb{E}[\partial s / \partial \theta] = \mathbb{E}[\ell'']$. For the second, note $\partial p / \partial \theta = p \cdot s$ (since $s = \partial \log p / \partial \theta$), so the second integral is $\int s^2 p dx = \mathbb{E}[s^2]$. Therefore:

$$\mathbb{E}[\ell''] + \mathbb{E}[s^2] = 0 \implies \boxed{\mathbb{E}[s^2] = -\mathbb{E}[\ell'']}.$$

3.5.4 Geometric interpretation

The identity says: *on average over the data, the square of the slope of the log-likelihood equals minus the curvature of the log-likelihood.*

- $\ell''(\theta) < 0$ at a maximum (concave down): $-\mathbb{E}[\ell''] > 0$ makes sense as a measure of information.
- Large curvature $|\ell''|$ (sharply peaked log-likelihood) means the score s varies steeply with θ , so data are highly informative about θ .
- Small curvature (flat log-likelihood) means the score is nearly flat — the likelihood barely responds to changes in θ , so data are weakly informative.

3.5.5 The pre-integral quantity: score squared

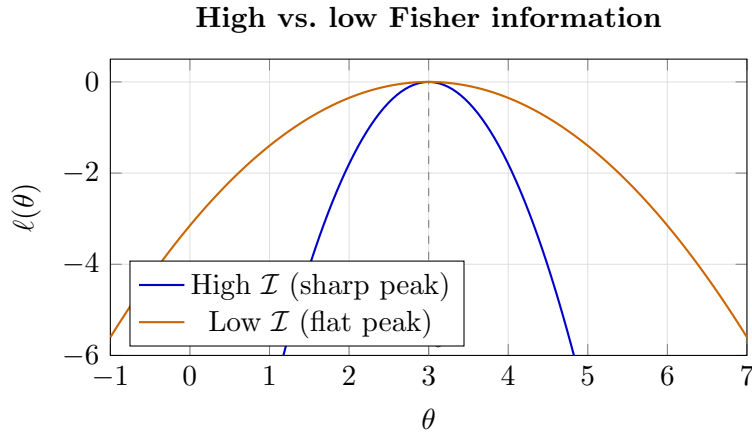
A subtle but important point: $\mathcal{I}(\theta) = \mathbb{E}[s(\theta)^2]$ is a number — an expectation. The quantity *under the integral* is $s(x, \theta)^2$, a **random variable** (it depends on x). This pre-integral object has no universally agreed name; one sometimes sees it called the *score contribution* or *observed information contribution*.

Its distribution is determined by the model: for $\mathcal{N}(\mu, \sigma^2)$ with $n = 1$,

$$s = \frac{x - \mu}{\sigma^2} \sim \mathcal{N}\left(0, \frac{1}{\sigma^2}\right), \quad s^2 \sim \frac{1}{\sigma^2} \chi_1^2.$$

The mean of s^2 equals $1/\sigma^2 = \mathcal{I}$ — this is the information equality by definition. But the distribution of s^2 is *right-skewed* (chi-squared); the Fisher information is just its expectation, not the full distribution.

Large $\mathcal{I}$: sharply curved log-likelihood, precise MLE. Small $\mathcal{I}$: flat log-likelihood, imprecise MLE.



3.6 Score properties: a compact reference

3.6.1 Two distinct zeros

The score $s(x, \theta)$ is a function of *two* arguments and appears in two different zero conditions that are often confused.

Zero mean under the true parameter (a property of the distribution):

$$\mathbb{E}_{\theta_0}[s(\cdot, \theta_0)] = 0.$$

Here x is random (drawn from $p(\cdot|\theta_0)$) and θ_0 is fixed. Both the sampling measure and the argument of s use the *same* θ_0 . Short version: *measure and score argument coincide.*

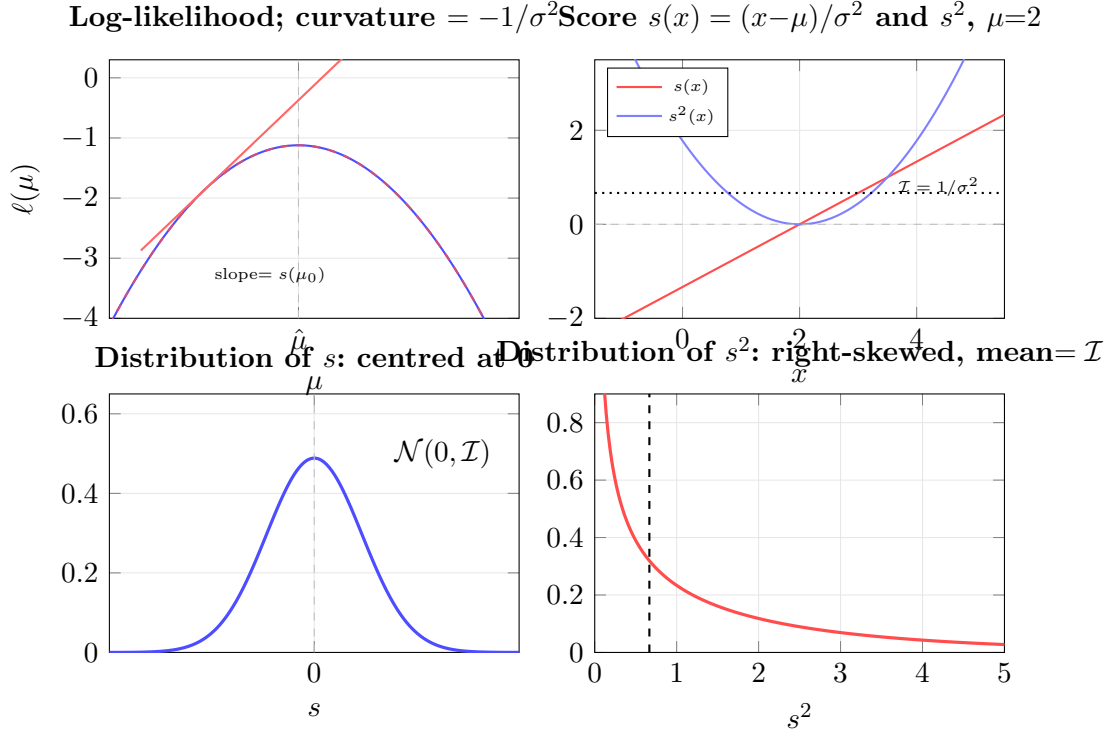


Figure 8: Four panels illustrating the information identity for $\mathcal{N}(\mu, \sigma^2)$ with $\sigma^2 = 1.5$, $\mu_{\text{true}} = 2$. Top-left: log-likelihood $\ell(\mu|x_1)$ with curvature $\ell_{\mu\mu} = -1/\sigma^2$ (osculating parabola, purple dashed) and a tangent at $\mu_0 \neq \hat{\mu}$ whose slope is the score. Top-right: score $s(x, \mu_{\text{true}}) = (x - \mu)/\sigma^2$ and s^2 as functions of x ; the horizontal dotted line marks $\mathcal{I} = E[s^2] = 1/\sigma^2$. Bottom-left: $\mathcal{N}(0, \mathcal{I})$ density (limiting distribution of s); centred at 0. Bottom-right: $(1/\sigma^2)\chi_1^2$ density (distribution of s^2); right-skewed; its mean (dashed) equals $\mathcal{I}$.

Zero at the MLE (a property of a specific dataset):

$$s(x_{\text{obs}}, \hat{\theta}) = 0.$$

Here x_{obs} is fixed (the observed sample) and θ varies. This is the first-order optimality condition; it requires ℓ to be differentiable and $\hat{\theta}$ to be in the interior of Θ (see Section 3.7).

What breaks centring. If the sampling measure is P_{θ_0} but the score is evaluated at a different $\theta_1 \neq \theta_0$:

$$\mathbb{E}_{\theta_0}[s(\cdot, \theta_1)] \neq 0 \quad \text{in general.}$$

The measure and the argument diverge; centring fails. This is exactly the situation under the alternative hypothesis H_1 .

3.6.2 LLN and CLT for the score (the Probabilist)

Score for a sample is a sum. For $n = 1$, the score is

$$s(x, \theta) = \frac{\partial_{\theta} f_{\theta}(x)}{f_{\theta}(x)}.$$

For an i.i.d. sample $(x_1, \dots, x_n)$ the joint density is $f_{\theta}(x_1) \cdots f_{\theta}(x_n)$, and by the product rule for D (Property 1 of Section 3.2):

$$s(x_1, \dots, x_n; \theta) = \frac{\partial_{\theta}[f_{\theta}(x_1) \cdots f_{\theta}(x_n)]}{f_{\theta}(x_1) \cdots f_{\theta}(x_n)} = \frac{\partial_{\theta} f_{\theta}(x_1)}{f_{\theta}(x_1)} + \cdots + \frac{\partial_{\theta} f_{\theta}(x_n)}{f_{\theta}(x_n)} = s(\theta, x_1) + \cdots + s(\theta, x_n).$$

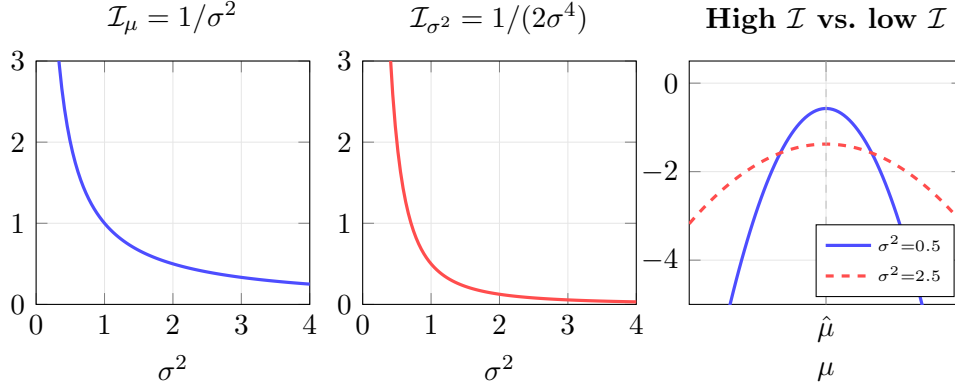


Figure 9: Fisher information for the normal model. Left: $\mathcal{I}_\mu = 1/\sigma^2$ as a function of σ^2 . Centre: $\mathcal{I}_{\sigma^2} = 1/(2\sigma^4)$ — faster decay; estimating variance is harder than estimating mean for large σ . Right: sharp (high $\mathcal{I}$) vs. flat (low $\mathcal{I}$) log-likelihood; same MLE $\hat{\mu} = x_1$.

Denoting $Y_i(\omega) := s(\theta, X_i(\omega))$: having i.i.d. X_i we get i.i.d. Y_i with zero mean (score identity), and

$$s(X_1, \dots, X_n; \theta) = Y_1 + \dots + Y_n, \quad \mathbb{E} s(X_1, \dots, X_n; \theta) = 0.$$

LLN: n is too much. Dividing by n :

$$\frac{1}{n} s(X_1, \dots, X_n; \theta) = \bar{Y}_n \xrightarrow{\text{a.s.}} \mathbb{E} Y_1 = 0.$$

Norming by n makes the score degenerate to zero.

CLT: divide by $\sqrt{n}$.

$$\frac{1}{\sqrt{n}} s(X_1, \dots, X_n; \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2), \quad \sigma^2 = \text{Var} Y_1. \quad (1)$$

By the information identity, that variance equals the Fisher information:

$$\mathcal{I}(\theta) = \text{Var} s(\theta, X_1) = \text{Var} Y_1.$$

Normalising to a standard pivot:

$$\frac{1}{\sqrt{n} \sqrt{\text{Var} s(X_1; \theta)}} (s(X_1; \theta) + \dots + s(X_n; \theta)) \xrightarrow{d} \mathcal{N}(0, 1). \quad (2)$$

CLT for squares: χ^2 limit. From the base CLT (2), convergence for squares:

$$\frac{1}{n \mathcal{I}_1(\theta)} s^2(X_1, \dots, X_n; \theta) \xrightarrow{d} \chi_1^2. \quad (3)$$

Note $n \cdot \mathcal{I}_1(\theta) = \mathcal{I}_n(\theta)$, so (3) can be written

$$\frac{s^2(X_1, \dots, X_n; \theta)}{\mathcal{I}_n(\theta)} \xrightarrow{d} \chi_1^2.$$

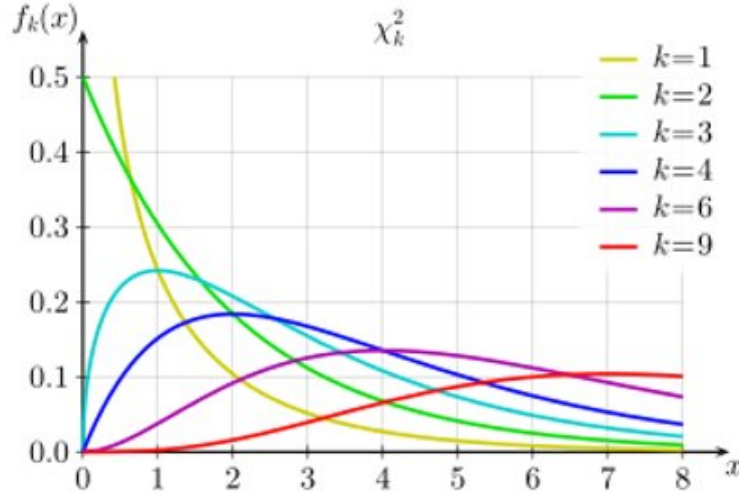


Figure 10: The χ_k^2 family for $k = 1, 2, 3, 4, 6, 9$. For $k = 1$ (yellow) the density is unbounded at the origin and strictly decreasing; for $k \geq 3$ a mode appears at $x = k - 2$. The score-squared statistic converges to χ_1^2 under H_0 .

3.6.3 The square of the CLT: diagonal and off-diagonal decomposition

We follow the Probabilist's notes of June 5, 2026.

The base CLT (2) gives a $\mathcal{N}(0, 1)$ limit for the normalised score. Squaring both sides of (2) gives the χ_1^2 limit (3). It is instructive to see *what carries that limit* inside the squared sum. We work with general i.i.d. $X_i \sim (0, \sigma^2)$; the score setting is the special case $X_i = s(\theta, \xi_i)$.

Start from

$$\frac{1}{\sigma\sqrt{n}} (X_1 + \cdots + X_n) \xrightarrow{d} N(0, 1).$$

Square both sides:

$$\frac{1}{\sigma^2 n} (X_1 + \cdots + X_n)^2 \xrightarrow{d} \chi_1^2.$$

Expand the square:

$$(X_1 + \cdots + X_n)^2 = \sum_{i=1}^n X_i^2 + \sum_{i \neq j} X_i X_j,$$

so

$$\underbrace{\frac{1}{\sigma^2 n} \sum_{i=1}^n X_i^2}_{\text{diagonal}} + \underbrace{\frac{1}{\sigma^2 n} \sum_{i \neq j} X_i X_j}_{\text{off-diagonal}} \xrightarrow{d} \chi_1^2.$$

Diagonal term: degenerates by LLN. Since $EX_i^2 = \sigma^2$, the LLN gives

$$\frac{1}{\sigma^2 n} \sum_{i=1}^n X_i^2 \xrightarrow{a.s.} 1.$$

Off-diagonal term: the source of limiting randomness. Define $T_n := \frac{1}{n} \sum_{i \neq j} X_i X_j$. Each term is centred: $EX_i X_j = EX_i \cdot EX_j = 0$, so $ET_n = 0$. The off-diagonal terms do *not* degenerate. Since the diagonal converges to 1, it is T_n/σ^2 that carries the entire χ_1^2 randomness:

$$\frac{T_n}{\sigma^2} = \frac{1}{\sigma^2 n} \sum_{i \neq j} X_i X_j \xrightarrow{d} \chi_1^2 - 1.$$

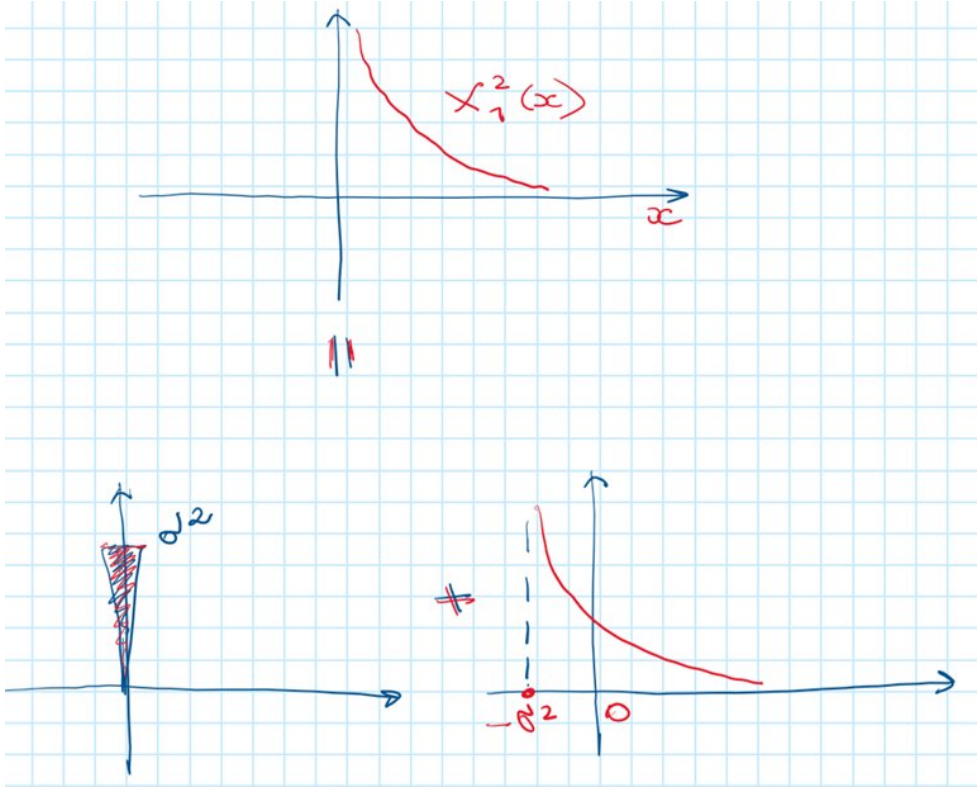


Figure 11: The Probabilist's diagram for the decomposition of the squared sum. *Top*: the χ_1^2 density. *Bottom left*: the diagonal term concentrates on σ^2 (hatched spike); *bottom right*: the off-diagonal term T_n/σ^2 is the source of the limiting randomness — it is supported on $(-\sigma^2, \infty)$ (the $\chi_1^2 - 1$ shift).

Variance of a single off-diagonal term

For $i \neq j$, using independence:

$$\text{Var}(X_i X_j) = E(X_i X_j)^2 - [E(X_i X_j)]^2 = E X_i^2 \cdot E X_j^2 = (\sigma^2)^2 = \sigma^4.$$

Covariance structure of the off-diagonal products

For $i \neq j$ and $k \neq l$:

All four indices distinct. $\{i, j\} \cap \{k, l\} = \emptyset$, all variables independent, each mean-zero, so $\text{Cov}(X_i X_j, X_k X_l) = E X_i \cdot E X_j \cdot E X_k \cdot E X_l = 0$.

Exactly one index shared. Representative: $\text{Cov}(X_1 X_2, X_2 X_3) = E X_1 X_2^2 X_3 = E X_1 \cdot E X_2^2 \cdot E X_3 = 0$, since $E X_1 = E X_3 = 0$.

Both indices shared. $\text{Cov}(X_i X_j, X_i X_j) = \text{Var}(X_i X_j) = \sigma^4$.

Summary:

$$\text{Cov}(X_i X_j, X_k X_l) = \begin{cases} \sigma^4 & \text{if } \{i, j\} = \{k, l\}, \\ 0 & \text{otherwise.} \end{cases}$$

Covariance matrix for $n = 2$

With $n = 2$, $X_1, X_2 \sim \text{iid}(0, \sigma^2)$:

$$\text{Cov} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \sigma^2 I, \quad \text{Cov} \begin{pmatrix} X_1 X_2 \\ X_2 X_1 \end{pmatrix} = \begin{pmatrix} \sigma^4 & \sigma^4 \\ \sigma^4 & \sigma^4 \end{pmatrix}, \quad E(X_1 X_2 + X_2 X_1) = 0.$$

Vector θ . When $\theta = (\theta_1, \dots, \theta_k)^\top \in \mathbb{R}^k$, the score is the gradient vector:

$$s(x, \theta_1, \dots, \theta_k) = \nabla_\theta \ell(x; \theta_1, \dots, \theta_k) = \left(\frac{\partial \ell}{\partial \theta_1}, \dots, \frac{\partial \ell}{\partial \theta_k} \right)^\top,$$

and the Fisher information matrix is

$$\mathcal{I}(\theta_1, \dots, \theta_k) = \mathbb{E}[s s^\top] = \begin{pmatrix} \mathbb{E}\left(\frac{\partial \ell}{\partial \theta_1}\right)^2 & \cdots & \mathbb{E}\left(\frac{\partial \ell}{\partial \theta_1}\right)\left(\frac{\partial \ell}{\partial \theta_k}\right) \\ \vdots & & \vdots \\ \mathbb{E}\left(\frac{\partial \ell}{\partial \theta_k}\right)\left(\frac{\partial \ell}{\partial \theta_1}\right) & \cdots & \mathbb{E}\left(\frac{\partial \ell}{\partial \theta_k}\right)^2 \end{pmatrix}.$$

The CLT for the vector score gives

$$s^\top \mathcal{I}_n^{-1} s \xrightarrow{d} \chi_k^2.$$

3.6.4 From centring to concentration: what else is needed

Centring ($\mathbb{E}[s] = 0$) alone does *not* imply that values of s concentrate near zero. A symmetric heavy-tailed distribution has mean zero but rarely produces small values. Three additional ingredients are needed.

(i) **Finite variance.** $\text{Var}_{\theta_0}[s(\cdot, \theta_0)] = \mathcal{I}(\theta_0) < \infty$. This is not automatic: it requires the model to satisfy the regularity conditions under which the information equality holds. When $\mathcal{I} < \infty$, most mass of s lies within a few multiples of $\sqrt{\mathcal{I}}$ from zero.

(ii) **Scaling up: the sample score.** For an i.i.d. sample of size n , the total score is $S_n = \sum_{i=1}^n s(x_i, \theta_0)$. Its mean is still zero, but its variance grows: $\text{Var}[S_n] = n\mathcal{I}$. The score grows as $\sqrt{n}$, not as n , which is why one normalises: $S_n/\sqrt{n\mathcal{I}}$.

(iii) **Asymptotic normality (CLT).** Under standard regularity conditions,

$$\frac{S_n}{\sqrt{n\mathcal{I}(\theta_0)}} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{under } H_0.$$

It is the CLT — not the definition of centring — that delivers unimodality and near-symmetry around zero. For any fixed n , the distribution of the normalised score may be skewed or multi-modal; normality is an asymptotic statement.

3.6.5 Connection to the score test

Under H_0 : the normalised score is approximately $\mathcal{N}(0, 1)$ — small values are typical.

Under H_1 : the sampling measure is P_{θ_1} but we evaluate the score at the null θ_0 . Measure and argument diverge, so $\mathbb{E}_{\theta_1}[s(\cdot, \theta_0)] \neq 0$: the score has a non-zero mean (a *location shift*). The larger the departure from H_0 , the further the score is shifted from zero.

The score (LM) test therefore rejects H_0 when the observed score at $\hat{\theta}_0$ is too far from zero relative to $\sqrt{\mathcal{I}}$. The intuition is: *if I were sampling from the null model, the score should be centred here; a large score signals I am not.*

3.7 When score is zero at the MLE: smooth vs non-smooth cases

The condition $s(\hat{\theta}) = 0$ requires: (a) ℓ differentiable at $\hat{\theta}$; (b) $\hat{\theta}$ interior to Θ . Both can fail.

Case 1: non-differentiable likelihood (Laplace). For $p(x | \theta) = \frac{1}{2}e^{-|x-\theta|}$, the log-likelihood $\ell(\theta) = -|x_1 - \theta| + \text{const}$ has a kink at $\theta = x_1$. The MLE is the argmax, but characterised by the subdifferential condition $0 \in \partial \ell(\hat{\theta})$, not by $s(\hat{\theta}) = 0$.

Object	Type	Key property
$s(x, \theta)$	random variable (in x , fixed θ)	$\mathbb{E}_\theta[s(\cdot, \theta)] = 0$
$s(x, \theta)^2$	random variable (pre-integral quantity)	$\mathbb{E}_\theta[s^2] = \mathcal{I}(\theta)$
$\mathcal{I}(\theta) = \mathbb{E}[s^2]$	number (parameter)	$= \text{Var}[s] = -\mathbb{E}[\ell'']$
$S_n = \sum_i s(x_i, \theta)$	random variable (n obs.)	$\mathbb{E}[S_n] = 0, \text{Var}[S_n] = n\mathcal{I}$
$S_n/\sqrt{n\mathcal{I}}$	normalised score	$\xrightarrow{d} \mathcal{N}(0, 1)$ under H_0

Table 1: Score objects and their key properties. In every row the expectation is taken under the *true* parameter θ_0 .

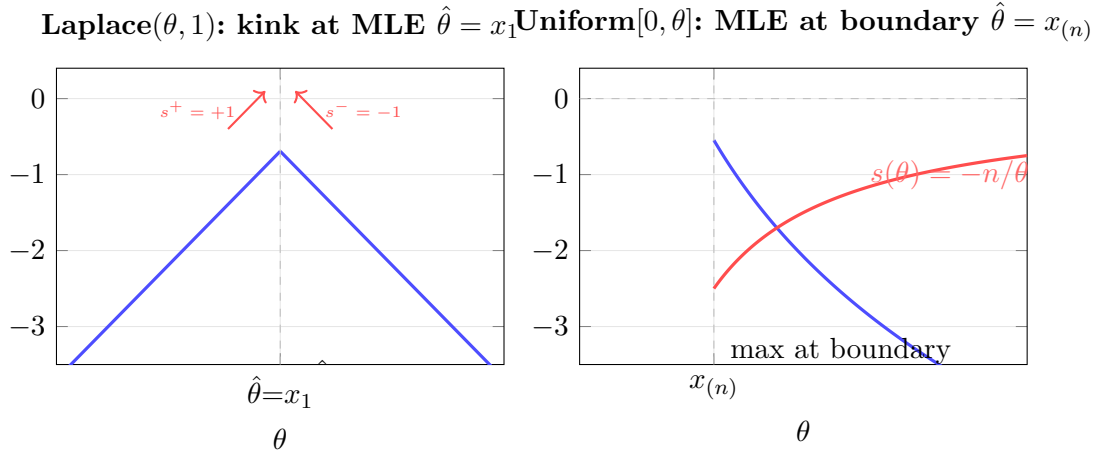


Figure 12: Non-smooth cases. Left: Laplace($\theta, 1$) — log-likelihood has a kink at the MLE; score does not exist there, but the MLE is well-defined. Left and right derivatives exist and have opposite signs: $0 \in \partial\ell(\hat{\theta})$ (subdifferential condition). Right: Uniform[$0, \theta$] — MLE at the boundary $\hat{\theta} = x_{(n)}$; score $s(\theta) = -n/\theta < 0$ everywhere on the interior, so no stationary point exists.

Case 2: boundary maximum (Uniform). For $X_i \sim \text{Uniform}[0, \theta]$, $\ell(\theta) = -n \log \theta$ for $\theta \geq x_{(n)}$. The score $s(\theta) = -n/\theta < 0$ is strictly negative on the interior: the log-likelihood is monotone decreasing, so the maximum is at the boundary $\hat{\theta} = x_{(n)}$. No stationarity condition applies.

Remark 8. The LM test theory assumes the regular smooth interior case. For non-smooth likelihoods, subgradient methods or boundary corrections are required.